

Performance and cost experiments

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Accessibility and user feedback

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Incident investigation

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Deployment environments

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Performance and cost experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Performance and cost experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

$C=[1, 2, 3, 10]$, $D=[3, 3, 3, 3]$. Compute means/medians; one D response is wrong.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Durations $A=[1, 1, 1, 9]$ seconds, $B=[2, 2, 2, 2]$. All pass. Compute mean and median. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Resource improvement under one workload is not universal efficiency or equal quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

$C=[1, 2, 3, 10]$, $D=[3, 3, 3, 3]$. Compute means/medians; one D response is wrong.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled performance and cost experiments evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of performance and cost experiments. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Performance and cost experiments

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

Mean resource use per case

	Seconds/case	Cost units/case
A	3	2
B	2	3

Read a cell using both its row and column labels.

Performance and cost experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute mean time

A averages 3 seconds; B averages 2 seconds per case.



2. Compute mean cost

A averages 2 units; B averages 3 units per case.



3. Compare trade-off

B saves one second but costs one extra unit per case.

CHANGE ONE THING

Require at most 2 cost units per case.

Explain the changed result using the matching lesson.

Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B saves one second but costs one extra unit per case.

1. Compute mean cost
2. Compute mean time
3. Compare trade-off

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare trade-off B saves one second but costs one extra unit per case. A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

Practice 2: explain the deciding evidence

Changed-case practice: Require at most 2 cost units per case. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. A meets the budget; B does not.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A meets the budget; B does not. The mandatory resource condition changes the decision.

Repair a missing building block

Review Accessibility and user feedback (ai-6/28). Goal: Explain the mechanism and terminology of accessibility and user feedback. Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?. Return to this week and explain what you changed.

Review Incident investigation (ai-6/26). Goal: Explain the mechanism and terminology of incident investigation. Short practice: Order the three incident records and locate the old evidence version. How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the mechanism and terminology of deployment environments. Short practice: Identify the staging data and credential boundary. Why must the rehearsal stop if its credential reaches production even though the process is labelled staging?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
